

“Critical slowing down in depression” is a great idea that still needs empirical proof

With great interest we read the paper by van de Leemput et al. (1). The authors conceptualize depression as a complex dynamic system with two distinct states (normal and depressed) and support this idea with dynamic system theory and model simulations. Unfortunately, when presenting empirical evidence for their supposition, the authors fall into the trap of generalizing group-level results to the individual level and mixing up between-subject and within-subject variability, a trap often encountered in the medical field (2).

The authors study depressed and non-depressed persons during a baseline mood-monitoring period and subsequent follow-up. Three indicators of critical slowing down (elevated autocorrelation, variance, and correlation between emotions) were associated with upcoming transitions. This evidence was based on between-subject variability: persons with higher autocorrelation, variance, and correlation compared with others also showed higher (change in) follow-up scores than others. Nevertheless, the authors interpret the evidence as a within-subject effect, using phrases like “increasing autocorrelation,” “rising variability,” and “critical slowing down.” This is questionable for two reasons. First, the design does not allow a within-subject interpretation, because the indicators of critical slowing down were not measured repeatedly within individuals. Second, higher autocorrelation, variance, and correlation may also be explained by higher baseline levels on the emotion variables. Autocorrelation was estimated

in a multilevel model using the interaction between an autoregressive parameter and follow-up depression score. Estimates in multilevel models, however, are a mixture of between- and within-effects, if these effects are not properly disaggregated (2, 3). The autoregressive parameter will then “pick up” variance due to between-subject differences in mean levels. Thus, the higher autocorrelation found in the high-risk group may actually reflect higher baseline levels of the emotions.

Variances were assessed by estimating random effect variances in a three-level model. However, the variance is known to be related to the mean and may be especially low in the case of floor and ceiling effects (4), as in the present study. Thus, also between-subject differences in variance may reflect differences in baseline levels of the emotions. The authors therefore rightly state that the variance may not be the best indicator of critical slowing down (which may also explain why they discount the contradictory evidence of a lower variance in the high-risk group in two of the emotions; their appendix). The correlations between the emotions were estimated by random effect correlations of the three-level model. However, correlation coefficients are affected by the variances of the emotions (5). Because these variances may be confounded by baseline levels of the emotions, the correlations among the emotions may be as well.

To summarize, we think that the supposed within-subject effect can just as well be ex-

plained by between-subject differences in baseline levels of the emotions. Further, whether the assumed “phase transition” really takes place is also uncertain; the change may just as well be gradual or minimal. Therefore, the authors’ conceptualization of depression as a dynamic system is appealing and may be right, but the empirical evidence is weak. Future studies should disaggregate inter- and intraindividual variability more carefully.

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- 1 van de Leemput IA, et al. (2014) Critical slowing down as early warning for the onset and termination of depression. *Proc Natl Acad Sci USA* 111(1):87–92.
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Reply to Bos and De Jonge: Between-subject data do provide first empirical support for critical slowing down in depression

With interest we read the response of Bos and De Jonge (1) to the paper by van de Leemput et al. (2). They argue that our paper (i) rests on illegitimate generalization of group level results to the individual level and (ii) statistically mixes up between-subject and within-subject variability.

With respect to i, we indeed provided support for an intraindividual model through individual differences data. The logic behind our analysis is simply that, if individual people display early warning signals when closing in on a transition, then individuals who are closer to a tipping point should show higher levels of autocorrelation and variance. In our data, the subsamples of individuals who proved to be in a different state on follow-up indeed showed such warning signals. Although the tested statistical hypothesis concerns the population distribution of autoregressive coefficients, as Bos and De Jonge note, that hypothesis is derived directly from a theory on the structure of intraindividual processes; this derivation was made consciously and with care, as testified by our supporting information. Thus, we did not mix up levels of analysis. However, we do agree with Bos and De Jonge that time series assessments obtained while individuals undergo a transition would be ideal, because these would allow for direct intraindividual tests of our hypothesis.

With respect to ii, the authors suggest that between- and within-effects were not properly disaggregated in our multilevel models for autocorrelation. This suggestion is incorrect. We used the strategy of person-mean centering, which unambiguously disaggregates these effects (3). As stated in our paper, the effect of the mean is indeed one of the

issues with analyzing variances (4); however, we addressed this issue through the analysis of the coefficients of variation (see ref. 2, supporting information). This analysis shows effects of the means but also reveals that the elevated variation in mood as an early warning of an upcoming depression in the general population is robust against correction for the mean. Thus, the data do not support the alternative explanation of the results as put forward by Bos and De Jonge. This defuses the argumentation underlying their conclusion that the evidence presented is weak.

We agree with the authors that we cannot exclude the possibility that individual transitions may have been gradual or minimal. Future within-subject designs may further elucidate the presence of relevant transitions in this regard. However, the current group-based findings were consistent with expectations following from the principles of dynamic systems, and our paper reports empirical results compatible with the idea that depression behaves as a complex system. Naturally, we are aware of the limitations of between-subject findings and strongly agree that future research should aim to examine intraindividual changes in early warning signals and whether these anticipate clinically relevant intraindividual transitions in depression.

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